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Roshan Kafle

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Nationality: Nepalese

ABOUT ME

Certified Geomatics Engineer and Backend/WebGIS Developer at NAXA where I build tools for AI-based Earth Observation, spatial data processing, and decision-support systems. Experienced in GeoAI, combining drone and Sentinel data and have worked with international projects like GeoNadir. Focused on applying AI with EO technologies to monitor land and environmental change, with a focus on sustainable solutions for the Global South. Outside of work, I enjoy hiking, meeting new people, sharing ideas, and collaborating across cultures.

EDUCATION AND TRAINING

Bachelor in Geomatics Engineering

Tribhuvan University [1 Oct 2018 – 22 Apr 2023]

City: Pokhara | **Country:** Nepal | **Website:** https://tu.edu.np/

Relevant courses: Remote Sensing, Photogrammetry, Geospatial Programming (Python, MATLAB, and C++), Geodesy, Satellite Positioning

High School (Faculty of Science)

SOS Hermann Gmeiner School [1 Jun 2016 – 1 Jun 2018]

City: Pokhara | **Country:** Nepal | **Website:** https://soshgsgandaki.edu.np/

PROJECTS

[1 May 2022 – 3 Apr 2023]

Precision Farming – Final Year Thesis

Developed an image classification model using TensorFlow and MobileNetV2 to detect nutrient deficiencies in tomato leaves, trained with 1,500+ labeled images. Used transfer learning, CNNs, and dropout layers to enhance model performance and reduce overfitting. The model was used for real-world deployment in smart agriculture.

Link: https://github.com/Kafle33/fastapi-tomato-classifier

WORK EXPERIENCE

GeoNadir – Queensland, Australia

Website: https://geonadir.com/

WebGIS/Backend Developer

[1 Jan 2024 – Current]

- Integrated Copernicus satellite data in GeoNadir's drone mapping platform for a globally accessible and collaborative EO system improving geospatial analysis and visualization.
- Applied AI-based classification using Object-Based Image Analysis (OBIA) and Pixel-Based Image Analysis (PBIA) on multi-band datasets for precise land cover mapping.
- Developed features using zero-shot image segmentation(Segment Anything Model & Grounding DINO) to digitize features.
- Predicted feature density in Sentinel-2 imagery from heatmaps of drone-classified raster data and linear regression.
- Performed geospatial processing and analysis using cloud computing platforms on AWS (S3, EC2, Lambda, RDS), ensuring performance, scalability, and secure data handling.
- Supported GeoNadir's FAIR data mission by enabling scalable, interoperable drone-to-satellite datasets with open access and reproducible pipelines.

NAXA Pvt. Ltd. – Kathmandu, Nepal

Website: https://naxa.com.np

WebGIS/Backend Developer

[1 Aug 2023 – 30 Dec 2023]

Khaderi-Alert (Drought Monitoring System)

- Integrated multi-source Earth Observation with meteorological data and automated scalable pipelines for real-time drought modelling and forecasting to support Humanitarian action.
- Developed a modular system for drought severity assessment (SPI, SPEI, PDSI), spatio-temporal hotspot mapping, and risk ranking to identify and track drought-prone areas for early warning and resource planning.
- Created an interactive Web-GIS platform with dynamic visualization, customizable map layers, real-time alerts, and capacity-building programs to empower stakeholders in effective drought management and decision-making.

AI & Urban Planning Analyst – DMAPS

- Used AI and Earth Observation data (aerial surveys, photogrammetry) to digitise urban landscapes and generate spatial models for resilient infrastructure planning.
- Engineered backend systems using PostGIS, optimising complex spatial queries with GIST indexing to handle large sized environmental datasets.
- Developed geospatial tools to support sustainable, interoperable urban planning and municipal service optimisation, including Janakpur's digital addressing (8,932 buildings) and Bheemdatta Municipality (36,139 buildings).

FMTM-HOTSOM (Open Source Contribution)

- Integrated geospatial data using Python and PostGIS, supporting data validation, storage, and seamless workflow from field collection to make final map products.
- Collaborated with international teams (HOT Tech Team, NAXA) to enhance the system's usability, reliability, and scalability for urban mapping and disaster mapping projects.

WebGIS/Backend Trainee

[1 May 2023 – 1 Aug 2023]

- Developed RESTful APIs using Django/FastAPI to serve geospatial data (GeoJSON, raster tiles) from PostGIS for real-time WebGIS applications.
- Integrated GeoServer with backend services to serve OGC-compliant layers (WMS/WFS), supporting interoperable digital mapping workflows.
- Implemented automated backend workflows for spatial data processing and tile generation, optimizing performance for large-scale geospatial applications.

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS [LINK]

[25 Jul 2025 – 26 Sep 2025]

Esri, Udemy, and AI Certifications

- Automated Sentinel-1 SAR workflows in ArcGIS Notebooks using Python for preprocessing, filtering, and feature extraction.
- Used deep learning methods in ArcGIS Pro for object detection and image classification with CNN and transfer learning.
- Did predictive modeling and spatiotemporal trend analysis of raster data in ArcGIS Online using harmonic regression and Mann-Kendall trend model.
- Completed Udemy courses on Deep Learning Applications for Earth Observation and Mastering Earth Observation with AI and Google Earth Engine, focused on AI-based EO analysis and Google Earth Engine automation.

[8 Apr 2023 – 11 Apr 2023]

DELTA 3.0 – Innovation Expo Presentation

- Applied convolutional neural networks (CNNs) on leaf imagery to detect nutrient deficiencies in tomato crops.
- Developed SmartAG tool as a WebGIS-integrated deep learning tool for crop nutrient classification.
- Demonstrated precision farming methods with crop health assessment and visualization to farmers.

[1 Nov 2022 – 10 Nov 2022]

Differential GPS (DGPS) and Terrestrial LiDAR Scanner Training

- Practical training on Differential GPS (DGPS) and Terrestrial LiDAR Scanner for high-resolution terrain mapping and accurate surveying.
- Collected, processed, and interpreted data using DGPS receiver, LiDAR scanner, and point cloud processing software for 3D model creation.
- Improved skills in using DGPS and LiDAR workflow for field geomatics work and spatial data analysis.

[1 Jun 2022 – 10 Jun 2022]

GNSS Applications and Technologies - Tokyo's CSIS & ICG

- Learned principles of GNSS, satellite constellations (GPS, GLONASS, Galileo, BeiDou, QZSS, IRNSS), positioning methods (Absolute, DGNSS, RTK, PPP), and error mitigation techniques.
- Field data collection with GNSS receivers, post-processing workflows, accuracy assessment, and integration with GIS, remote sensing, and geodetic reference frames (ITRF, WGS84).
- Emerging trends: multi-frequency and multi-constellation systems, GNSS–IMU integration, and applications in autonomous systems, precision mapping, and disaster management.

SKILLS AND ABILITIES

Digital Skills

- **Geospatial Technologies**: WebGIS, Remote Sensing, GIS, UAV, GNSS, GEE
- **Web Development & Programming**: Python, Django, FastAPI, Flask
- **Cloud & DevOps**: Amazon Web Services (AWS): EC2, ECS, S3, Lambda, RDS
- **Databases**: PostgreSQL, MySQL, PostGIS
- **Containerization**: Docker, Docker Hub
- **Version Control**: GitHub, Git
- **Geospatial Software**: ArcGIS, QGIS, ENVI, OpenStreetMap, Agisoft, Geoserver
- **Operating System**: Linux, Windows, Mac

Leadership and Mentorship

- Led a backend team of more than 20 developers, coordinated with stakeholders, and organized biweekly meetings for work alignment.
- Guided junior engineers, gave presentations on GIS and new technologies, and represented the team in job fair and career counseling programs.
- Participated and led a team in hackathon to develop field mapping and task splitting algorithm.
- Helped in local health and community programs, including voluntary blood donation events organized by Red Cross.
- Contributed to Humanitarian OpenStreetMap (HOT) campaigns by mapping buildings and roads for disaster preparedness and open geospatial data.
- Volunteered as a mentor in a 3-month mentorship program organized by GESAN Community Club.
- Participated as mentor in GeoFest 2024 organized by Women in Geospatial Nepal, supporting gender diversity and inclusion.

LANGUAGE SKILLS

Mother tongue(s): Nepali

Other language(s): Hindi | English